TITLE: **The CLT Zoo: sixteen distributions, where the Law of Large Numbers and the Central Limit Theorem hold, and how far the normal approximation is for discrete sums**

SUBJECT: Aim, method and results of the analyses in version 2 of the `CLT-Zoo-exercise` repository

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1. Aim and scope

Version 1 of this project is a notebook with sixteen distributions (eleven continuous, five discrete), each with its parameter sliders and a preset. Its introduction and README describe a study of the Law of Large Numbers (LLN) and the Central Limit Theorem (CLT): a running mean, the sampling distribution of the mean for a sample size N and M Monte Carlo replications, the standardized mean against $\mathcal{N}(0,1)$, and normal curves that are switched off when the mean or the variance is infinite. The code does not contain any of these. It is a distribution explorer: for one distribution it draws the density or probability function, a sample, and a distribution function obtained by integrating the density numerically. Version 2 keeps the zoo as it is (names, parameters, slider ranges, presets, samplers, seed 42), corrects the explorer, adds the LLN and CLT views the README described, and asks three questions:

1. **Is the zoo correct, and what did the explorer of version 1 compute?** Analysis 1.
2. **Where on the sliders does the LLN hold, and how fast?** The Pareto slider alone covers the three regimes. Analysis 2.
3. **How close is the standardized mean to $\mathcal{N}(0,1)$ across the zoo, and exactly for the five discrete members?** Analysis 3.

2. The zoo and its moments

For a distribution with mean μ and variance σ^2, the skewness is $\gamma = E(X-\mu)^3/\sigma^3$ and the excess kurtosis $\kappa = E(X-\mu)^4/\sigma^4 - 3$. The *moment order* r^* is the number with $E|X|^r < \infty$ exactly for $r < r^*$: $r^* = \infty$ for every member of the zoo except Student-t ($r^* = \nu$), Pareto ($r^* = \alpha$) and F ($r^* = d_2/2$). The mean exists if $r^* > 1$, the variance if $r^* > 2$. The samplers are those of version 1 (`numpy.random.Generator`); Table 1 gives the parametrization they use.

Table 1. The sixteen distributions of version 1: parameters (names of version 1), mean and variance. Geometric counts trials up to the first success $(1, 2, \dots)$; Negative Binomial counts failures before the r-th success $(0, 1, \dots)$; Pareto has $P(X > x) = (x_m/x)^\alpha$ for $x \geq x_m$.

Distribution	Mean	Variance
Normal(μ, σ)	μ	σ^2
Uniform(a, b)	$(a+b)/2$	$(b-a)^2/12$
Exponential(scale s)	s	s^2
Gamma(k, θ)	$k\theta$	$k\theta^2$
Beta(α, β)	$\dfrac{\alpha}{\alpha+\beta}$	$\dfrac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$
Lognormal(m, s)	$e^{m+s^2/2}$	$(e^{s^2}-1)e^{2m+s^2}$
Laplace(μ, b)	μ	$2b^2$
Student-t(ν)	$0 \quad (\nu > 1)$	$\nu/(\nu-2) \quad (\nu > 2)$
Pareto(x_m, α)	$\alpha x_m/(\alpha-1) \quad (\alpha > 1)$	$x_m^2\alpha/[(\alpha-1)^2(\alpha-2)] \quad (\alpha > 2)$
Chi-square(k)	k	$2k$
F(d_1, d_2)	$d_2/(d_2-2) \quad (d_2 > 2)$	$\dfrac{2d_2^2(d_1+d_2-2)}{d_1(d_2-2)^2(d_2-4)} \quad (d_2 > 4)$
Bernoulli(p), Binomial(n, p)	$p, \;\; np$	$p(1-p), \;\; np(1-p)$
Poisson(λ)	λ	λ
Geometric(p), NegBinomial(r, p)	$1/p, \;\; r(1-p)/p$	$(1-p)/p^2, \;\; r(1-p)/p^2$

3. Theory and methods

LLN and CLT. Let $X_1, X_2, \dots$ be independent copies of X, $S_n = X_1 + \cdots + X_n$ and $\bar{X}_n = S_n/n$. If $E|X| < \infty$, then $\bar{X}_n \to \mu$ (Kolmogorov, 1933). If also $\sigma^2 < \infty$, then $Z_n = \sqrt{n}\,(\bar{X}_n - \mu)/\sigma \xrightarrow{d} \mathcal{N}(0, 1)$ (Lindeberg, 1922; Lévy, 1937), so that $|\bar{X}_n - \mu|$ is of order $n^{-1/2}$.

Without a finite variance (Gnedenko and Kolmogorov, 1954; Feller, 1971). For Pareto$(1, \alpha)$ the tail $P(X > x) = x^{-\alpha}$ puts X in the domain of attraction of a stable law of index α when $\alpha < 2$. For $1 < \alpha < 2$, $n^{1-1/\alpha}(\bar{X}_n - \mu)$ converges to a stable law, so $|\bar{X}_n - \mu|$ is of order $n^{1/\alpha - 1}$, slower than $n^{-1/2}$. For $\alpha < 1$ the mean is infinite and $S_n/n^{1/\alpha} = n^{1-1/\alpha}\bar{X}_n$ converges to a positive stable law, so $\bar{X}_n$ grows like $n^{1/\alpha - 1}$. Analysis 2 measures these exponents: $-1/2$ ($\alpha = 2.5$), $-1/3$ ($\alpha = 1.5$) and $+1/4$ ($\alpha = 0.8$).

Exact law of the sum. For nine members S_N has a closed form, which makes the distribution of Z_N exact:

$$\begin{aligned}
&\text{Normal}: \ \mathcal{N}(N\mu, N\sigma^2); \quad \text{Exponential}(s): \ \text{Gamma}(N, s);\\
&\text{Gamma}(k, \theta): \ \text{Gamma}(Nk, \theta); \quad \chi^2_k: \ \chi^2_{Nk};\\
&\text{Bernoulli}(p): \ \text{Bin}(N, p); \quad \text{Bin}(n, p): \ \text{Bin}(Nn, p);\\
&\text{Poisson}(\lambda): \ \text{Poisson}(N\lambda);\\
&\text{Geometric}(p): \ N + \text{NegBin}(N, p); \quad \text{NegBin}(r, p): \ \text{NegBin}(Nr, p).
\end{aligned} \tag{1}$$

Distance to the normal law. With $F_N(x) = P(Z_N \leq x)$ and Φ the standard normal distribution function, the Kolmogorov distance is $D_N = \sup_x |F_N(x) - \Phi(x)|$. For an integer-valued S_N, F_N is constant between the points $z_k = (k - N\mu)/(\sigma\sqrt{N})$, so the supremum is attained

at some z_k, either at the point or just before it:

$$D_N = \max_k \max\big(|P(S_N \le k) - \Phi(z_k)|, \\ |P(S_N \le k-1) - \Phi(z_k)|\big), \tag{2}$$

over every k that carries probability (all but 10^{-15} in each tail). For a continuous S_N, the supremum is located on a grid of step 0.001 on $[-10, 10]$ and refined by bounded scalar optimization. The *continuity correction* approximates $P(S_N \le k)$ by Φ at the midpoint between two lattice values; its error is

$$E_N = \max_k \Big| P(S_N \le k) - \Phi\Big(\frac{k + \frac{1}{2} - N\mu}{\sigma\sqrt{N}}\Big)\Big|. \tag{3}$$

Why the correction helps, and by how much. For an integer-valued X whose values have greatest common step 1 (true for all five discrete presets), the lattice Edgeworth expansion (Esseen, 1945; Gnedenko and Kolmogorov, 1954) is

$$F_N(x) - \Phi(x) = \Big[-\frac{\gamma}{6}(x^2-1) + \frac{1}{\sigma}\,Q\big(N\mu + \sigma\sqrt{N}x\big)\Big]\frac{\varphi(x)}{\sqrt{N}} + o\big(N^{-1/2}\big), \\ Q(t) = \lfloor t \rfloor - t + \tfrac{1}{2}, \tag{4}$$

where φ is the standard normal density. The first term is the skewness correction of continuous distributions; the second, a saw-tooth with values in $[-\frac{1}{2}, \frac{1}{2}]$, is the lattice term: it takes both extreme values next to every jump of F_N and is zero at the midpoints, which is why the continuity correction removes it. Both terms are largest in absolute value at $x = 0$, which gives the leading constants

$$\sqrt{N}\,D_N \to c_D = \varphi(0)\Big(\frac{1}{2\sigma} + \frac{|\gamma|}{6}\Big), \qquad \sqrt{N}\,E_N \to c_E = \varphi(0)\,\frac{|\gamma|}{6}, \\ \varphi(0) = 1/\sqrt{2\pi}. \tag{5}$$

For symmetric members ($\gamma = 0$) $c_E = 0$ and E_N is of order $1/N$. Analysis 3 checks (5) against the exact distances.

Simulation. Sample means are computed from M rows of N draws. The distance of M simulated values of Z_N to Φ contains noise; for $M =$ 20,000 its mean, when Z_N is exactly $\mathcal{N}(0,1)$, is 0.0061 (exact distribution of the Kolmogorov statistic).

From the equations to the code. Each step is a function in `cltzoo/` and is checked in `tests/run_tests.py` against an independent computation (the code of version 1 read from the legacy notebook, numerical integration of the densities, N-fold convolution, goodness-of-fit tests of the samplers, brute-force search on a fine grid):

Step	Equations	Code
Zoo, moments, r^*	Table 1	`ZOO[name].sample / .frozen / .moments / .moment_order`
Law of S_N	(1)	`ZOO[name].sum_law(N)`
D_N, E_N	(2), (3)	`kolmogorov_distance`, `corrected_error`
Lattice constants	(5)	`lattice_constants`
Running mean, sample means	—	`running_mean`, `running_mean_quantiles`, `sample_means`
Notebook views	—	`panels.explorer_figure`, `panels.clt_figure`, `widgets`

4. Analysis 1: The zoo and the explorer of version 1 (Figure 1)

Why. Every later result rests on the distributions being what they claim to be, and on knowing where their moments exist.

Method. The samplers, densities and moments of version 2 are compared with version 1 and with independent computations (unit tests). The code of version 1 is then run on its own presets: its distribution function, the integral of the density over the plotting window rescaled to end at 1, is compared with the exact one. Its density or probability function is also evaluated at every slider position of every parameter, the other parameters at their presets.

How to read the figure. Each panel: a sample of seed 42 (bars) and the exact density or probability function (line or dots), at the preset of version 1.

Finding. The sixteen samplers of version 1 are correct: version 2 reproduces their draws exactly, and each agrees with its exact law (Kolmogorov–Smirnov or chi-square test, 200,000 draws). Its densities and probability functions are exact. Its distribution function is not: the largest error is 0.022 for Pareto, where the density jumps at x_m, and 0.0093 for Uniform, whose window ran from -7.2 to 7.2 for a support of $[0, 1]$; for the other presets it is at most 0.0027. The probability functions of Binomial and Negative Binomial stopped with an `OverflowError` at 30 of 299 and 46 of 179 slider positions ($n \geq 171$; $r \geq 45$ or $p \leq 0.10$): the binomial coefficients were computed from `math.gamma`, which overflows for arguments above 171.6. The exact functions of version 2 are finite at all 2,844 valid slider positions (0 failures). At 92 of 302 positions of the Uniform sliders $b \leq a$, and version 1 replaced b silently by $a + 10^{-6}$. The mean does not exist at 3 of the 93 Pareto positions, 1 of the 80 Student-t positions and 2 of the 160 positions of d_2 for F; the variance at 13, 2 and 4 of them.

Interpretation. The zoo itself needed no change. Version 2 uses exact distribution functions from `scipy.stats`, reports an impossible Uniform range instead of altering it, and knows where the mean and the variance exist, which the LLN and CLT views need.

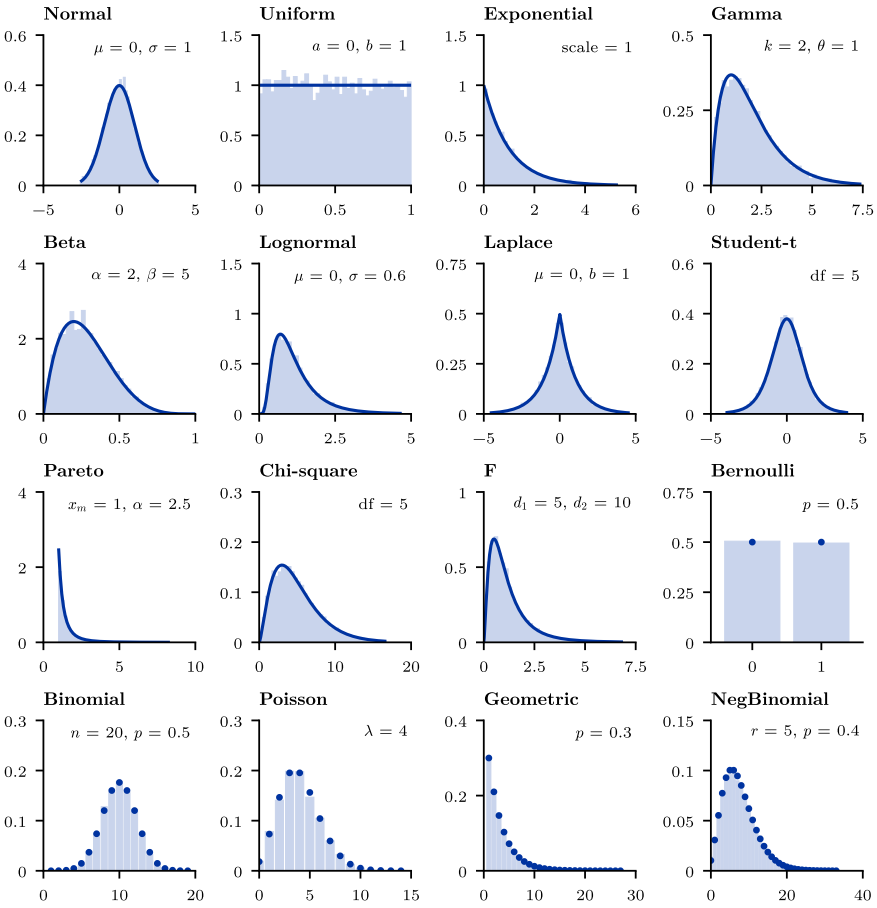


Figure 1. The zoo at the presets of version 1: exact density or probability function and a sample of seed 42 (the preset sample size of version 1, or 6,000 draws where its preset showed no sample).

5. Analysis 2: The LLN on the Pareto slider (Figure 2)

Why. The README of version 1 promised a running mean and normal curves that switch off when the mean or variance is infinite. Its Pareto slider (α from 0.8 to 10) contains all three regimes of Section 3, so the theory can be checked where the notebook lets the user go.

Method. Pareto$(1, \alpha)$ for $\alpha = 2.5$ (the preset), 1.5 and 0.8 (the smallest slider value). One running mean of 10^5 draws each (seed 42); then, over 4,000 independent paths, the median of $|\bar{X}_n - \mu|$ ($\alpha = 2.5, 1.5$) or of $\bar{X}_n$ ($\alpha = 0.8$) at 21 values of n from 10 to 10^5. The slope of its logarithm against $\log n$ is fitted for n from 10^3 to 10^5 and in each decade of n.

How to read the figure. (a) Dotted lines: μ. (b) Log–log scale: a straight line of slope s means a power n^s; the band holds the middle half of the paths; dotted lines have the theoretical slope.

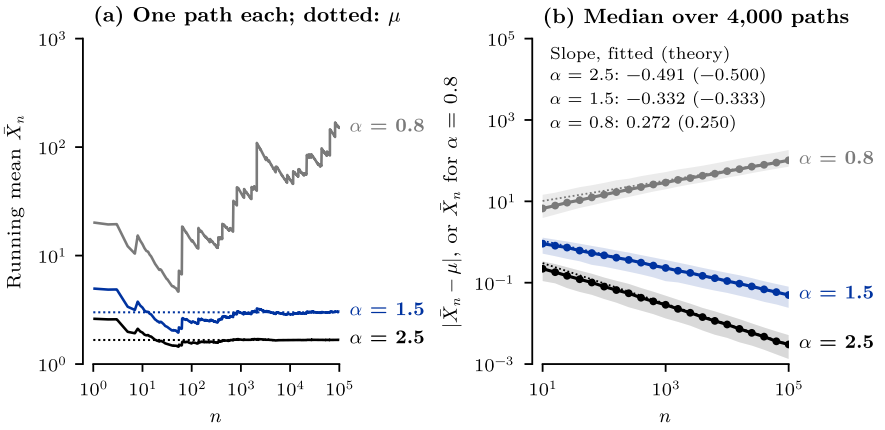


Figure 2. The LLN for Pareto$(1, \alpha)$: (a) one running mean for each α; (b) median distance to μ (or median running mean when $\mu = \infty$) over 4,000 paths, with quartiles.

Finding. The fitted slopes are -0.491 ($\alpha = 2.5$, theory -0.500), -0.332 ($\alpha = 1.5$, theory -0.333) and 0.272 ($\alpha = 0.8$, theory 0.250). The local slopes move toward the theoretical values as n grows: from -0.435 (n from 10 to 100) to -0.495 (10^4 to 10^5) for $\alpha = 2.5$, and from 0.336 to 0.267 for $\alpha = 0.8$; for $\alpha = 1.5$ they lie between -0.294 and -0.346. At $n = 10^5$ the median of $|\bar{X}_n - \mu|$ for $\alpha = 2.5$ is 0.00303, against $0.6745\,\sigma/\sqrt{n} = 0.00318$ from the CLT. For $\alpha = 0.8$ the single path ends at 150 and the median running mean is 102.

Interpretation. With a finite variance the error falls like $n^{-1/2}$; with $1 < \alpha \leq 2$ the mean is still estimated consistently, but at the slower rate $n^{1/\alpha - 1}$, and no normal curve describes the error; with $\alpha \leq 1$ there is nothing to converge to. Exponents measured on finite paths approach their limits only slowly, as the local slopes show. This is the behaviour the notebook now shows: in the LLN and CLT view the line for μ is drawn only when μ is finite, and the normal curves only when σ^2 is finite.

6. Analysis 3: The CLT across the zoo, and exactly for discrete sums (Table 2, Figures 3, 4)

Why. The README of version 1 promised the standardized mean against $\mathcal{N}(0, 1)$ for every member. For the five discrete members the distribution function of Z_N jumps, which sets a floor of order $1/\sqrt{N}$ on its distance to any continuous law; by (1) it can be computed exactly.

Method. (A) Every preset, $N = 30$: 20,000 sample means (seed 42) and the distance of their Z_{30} to Φ; for the nine members of (1) also the exact D_{30}. (B) The five discrete presets: exact D_N (2) and E_N (3) for N from 1 to 10^4, compared with (5); and the smallest N with $D_N < 0.01$

and with $E_N < 0.01$, searched up to $N = 3{,}000$.

How to read the figures. Figure 3: histograms of Z_{30} with the $\mathcal{N}(0,1)$ density; for discrete members each bar covers whole lattice values. Figure 4: log–log scale; dotted lines are $c/\sqrt{N}$ with c from (5); the red dotted line is 0.01.

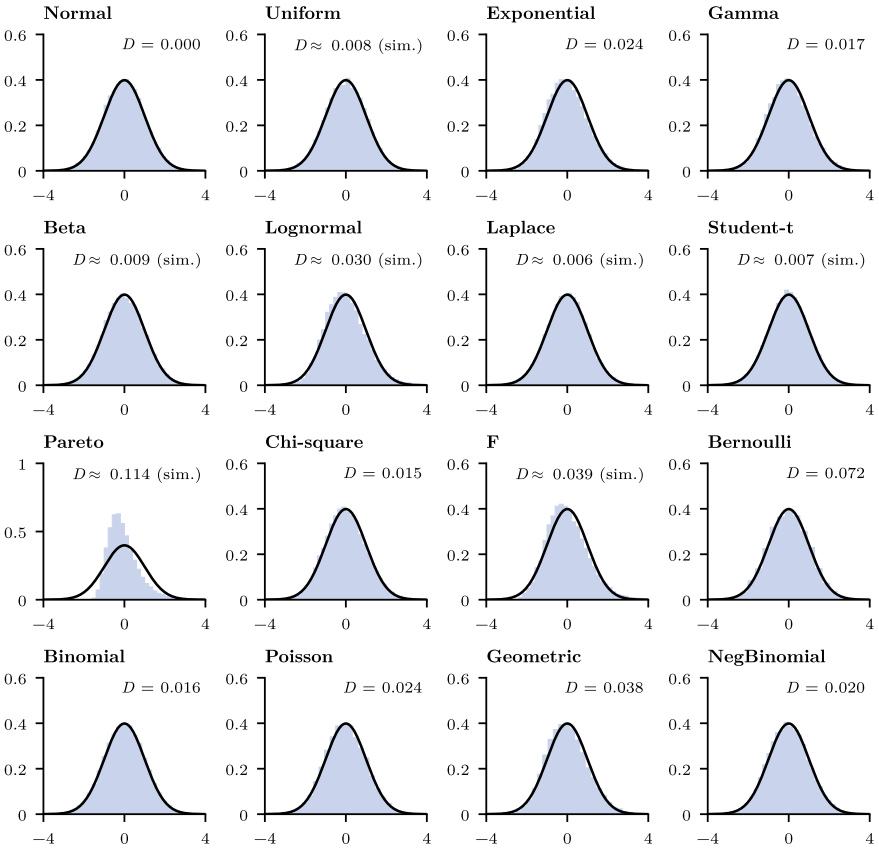


Figure 3. Standardized mean Z_{30} at every preset (20,000 sample means, seed 42) with the $\mathcal{N}(0,1)$ density, and the distance D to $\mathcal{N}(0,1)$: exact where the law of S_{30} is known, otherwise from the simulation.

Finding. At $N = 30$ the largest distances belong to Pareto (0.114, no finite third moment) and Bernoulli (0.072, exact; its standardized mean takes only 31 values), followed by Geometric (0.039), F (0.039) and Lognormal (0.030): the members with heavy or strongly skewed tails. Where the exact value is known the simulated distance agrees with it up to the noise level (Exponential: 0.024 exact, 0.027 simulated). For the five discrete presets $\sqrt{N}D_N$ and $\sqrt{N}E_N$ at $N = 10^4$ equal the constants of (5) to four decimals (Table 2). Without the correction, $D_N < 0.01$ requires $N = 1{,}592$ for Bernoulli and between 80 and 427 for the others; with it, $N = 4$, 1, 11, 38 and 183 suffice (Bernoulli, Binomial, Poisson, Negative Binomial, Geometric). For the symmetric

Table 2. (Top) $N = 30$: skewness $\gamma/\sqrt{30}$ of Z_{30}, exact D_{30} where available, and D from 20,000 simulated values (noise level 0.0061). (Bottom) Discrete presets: σ, γ, the constants (5) with $\sqrt{N}D_N$ and $\sqrt{N}E_N$ at $N = 10^4$, and the smallest N with $D_N < 0.01$ and with $E_N < 0.01$ (both stay below 0.01 for every larger N up to 3,000).

	$\gamma/\sqrt{30}$	exact	sim.
Normal	0.000	0.000	0.007
Uniform	0.000		0.008
Exponential	0.365	0.024	0.027
Gamma	0.258	0.017	0.018
Beta	0.109		0.009
Lognormal	0.413		0.030
Laplace	0.000		0.006
Student-t	0.000		0.007
Pareto	–		0.114
Chi-square	0.231	0.015	0.016
F	0.706		0.039
Bernoulli	0.000	0.072	0.074
Binomial	0.000	0.016	0.020
Poisson	0.091	0.024	0.030
Geometric	0.371	0.038	0.039
NegBinomial	0.169	0.020	0.019

	σ	γ	c_D	$\sqrt{N}D_N$	c_E	$\sqrt{N}E_N$	N_D	N_E
Bernoulli	0.500	0.000	0.3989	0.3989	0.0000	0.0003	1,592	4
Binomial	2.236	0.000	0.0892	0.0892	0.0000	0.0000	80	1
Poisson	2.000	0.500	0.1330	0.1330	0.0332	0.0332	177	11
Geometric	2.789	2.032	0.2066	0.2066	0.1351	0.1351	427	183
NegBinomial	4.330	0.924	0.1075	0.1075	0.0614	0.0614	116	38

members NE_N settles at 0.0272 (Bernoulli) and 0.0014 (Binomial), so E_N falls like $1/N$.

Interpretation. For a discrete population the error of the normal approximation has two parts of the same order $1/\sqrt{N}$: the jumps of the lattice and the skewness. The continuity correction removes the first exactly and leaves the second, so it is most useful for symmetric distributions (Bernoulli with $p = 0.5$: $N = 4$ instead of 1,592) and least for skewed ones (Geometric: 183 instead of 427). A distance of 0.01 is reached at very different N across the zoo, so a single rule such as $N \geq 30$ does not describe it.

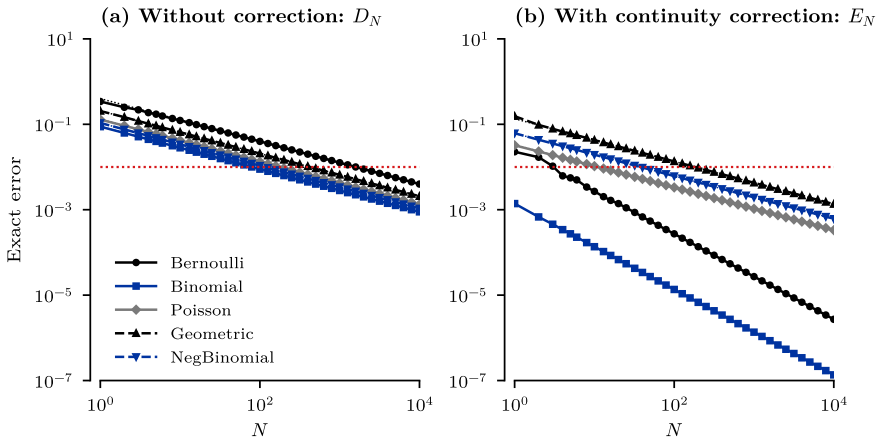


Figure 4. Discrete presets: exact distance of Z_N to $\mathcal{N}(0,1)$, (a) without and (b) with the continuity correction. Dotted: $c/\sqrt{N}$ from the lattice Edgeworth expansion.

7. Conclusions and limitations

- The zoo of version 1 is correct: samplers and densities are reproduced exactly. Its distribution functions were numerical (error up to 0.022), Binomial and Negative Binomial failed at 30 and 46 slider positions, and an impossible Uniform range was changed silently. Version 2 fixes all three and adds the LLN and CLT views that its README described.
- LLN: on the Pareto slider the error of the mean falls like $n^{-1/2}$, like $n^{1/\alpha-1}$ with infinite variance, and the running mean diverges for $\alpha \leq 1$; the measured exponents agree with theory to within 0.03 and approach it as n grows.
- CLT for discrete sums: the exact distances follow the lattice Edgeworth constants; the continuity correction removes the lattice term and leaves the skewness term, which is why it helps symmetric distributions most.
- Limitations: one parameter value per member except Pareto; the distances of the six continuous members without a closed-form sum are simulated, so below about 0.0061 they cannot be resolved; exact distances for continuous members are the subject of the CLT speed race project of the same author.

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NumPy and SciPy documentation: `numpy.random.Generator`, `scipy.stats`.

All numbers are generated from `figures/*.csv` by `make_paper.py`.